

Euclid's Elements — Book I

Proposition 29

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

Angle relations formed by a transversal of parallel lines

1 Introduction

Proposition I.29 is the first proposition in Book I whose proof genuinely uses Euclidean parallelism. Propositions I.27 and I.28 establish neutral implications of the form

angle conditions $\longrightarrow$ parallel lines,

whereas I.29 reverses the direction:

parallel lines $\longrightarrow$ angle relations.

For a transversal meeting two parallel lines, Euclid records three conclusions: alternate interior angles are equal, the exterior angle equals the interior and opposite angle, and the interior angles on the same side are together equal to two right angles.

The current Lean development makes a useful distinction. The genuinely Euclidean content is concentrated in the first, alternate-angle conclusion. The corresponding-angle and same-side conclusions are then neutral consequences of that result.

2 Classical Configuration

Let the transversal EF meet the parallel lines AB and CD at G and H , respectively:

$$A - G - B, \quad C - H - D, \quad E - G - H - F,$$

with

$$AB \parallel CD.$$

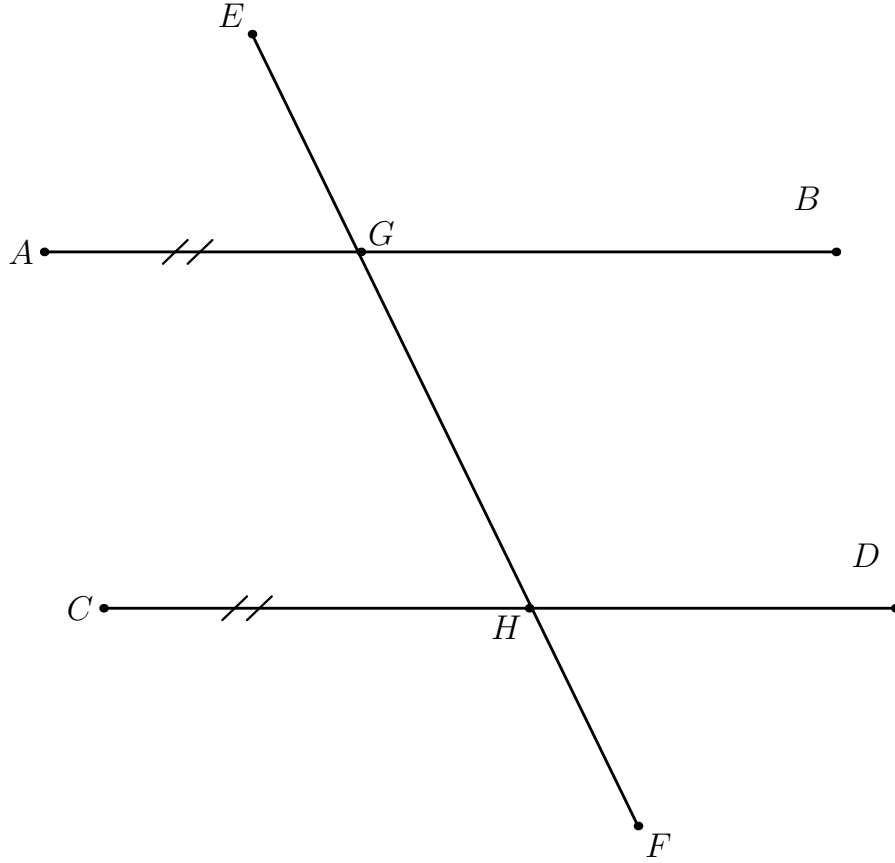


Figure 1: The classical transversal configuration for Proposition I.29. The lines AB and CD are given parallel; EF is the transversal. This single configuration supports all three conclusions of the proposition.

3 Euclid's Proof Architecture

Euclid's proof is not the same as the Hilbert reconstruction used by the project. In the classical argument the Fifth Postulate is used through the same-side interior-angle condition. Together with Proposition I.13 it rules out an inequality between the relevant alternate angles: if one were larger, the resulting pair of same-side interior angles would be less than two right angles, forcing the two supposedly parallel lines to meet on that side. Thus the alternate angles are equal.

Schematically, the classical core is

$$\begin{array}{c}
 AB \parallel CD \\
 \downarrow \\
 \text{assume the alternate angles are unequal} \\
 \downarrow \\
 \text{I.13 converts the inequality into a same-side sum} < 2R \\
 \downarrow \text{ Fifth Postulate} \\
 \text{the two lines would meet} \\
 \downarrow \\
 \perp \\
 \downarrow \\
 \angle AGH \cong \angle GHD.
 \end{array}$$

The remaining classical conclusions use only the same configuration. Vertical angles give the

corresponding-angle statement, while the straight angle at G supplies the supplementary relation needed for the same-side interior angles.

4 Current Lean Architecture

The production file imports only

`CGJteamLab.HilbertInterface`

and contains five public theorems.

4.1 Interface-level wrapper

The theorem

`euclid_proposition_29`

is a thin wrapper over

`equal_angles_from_parallel`

for the interface-level six-point configuration. The reusable theorem in turn delegates to the Euclidean alternate-angle theorem in the Hilbert interface. Thus the construction and the use of Group IV are no longer implemented locally in `Proposition29.lean`.

4.2 Explicit transversal theorem

The theorem

`euclid_proposition_29_transversal`

uses the explicit configuration

$$A - G - B, \quad C - H - D,$$

an incidence line `trans` through G and H , the orientation hypothesis

$$\text{OppositeSide}(A, D; \text{trans}),$$

and the given parallelism

$$AB \parallel CD.$$

It proves

$$\angle AGH \cong \angle GHD.$$

The proof first transports the global parallel relation to the local angle arms:

$$AB \parallel CD \longrightarrow GA \parallel HD.$$

This transport changes representation but not geometry.

The first genuinely new formal object is an interior point M chosen with

$$G - M - H.$$

The line-level Hilbert theorem then yields a congruence with M as the transversal-arm endpoint. Same-ray transport at G and H converts that raw result into the requested angles $\angle AGH$ and $\angle GHD$.

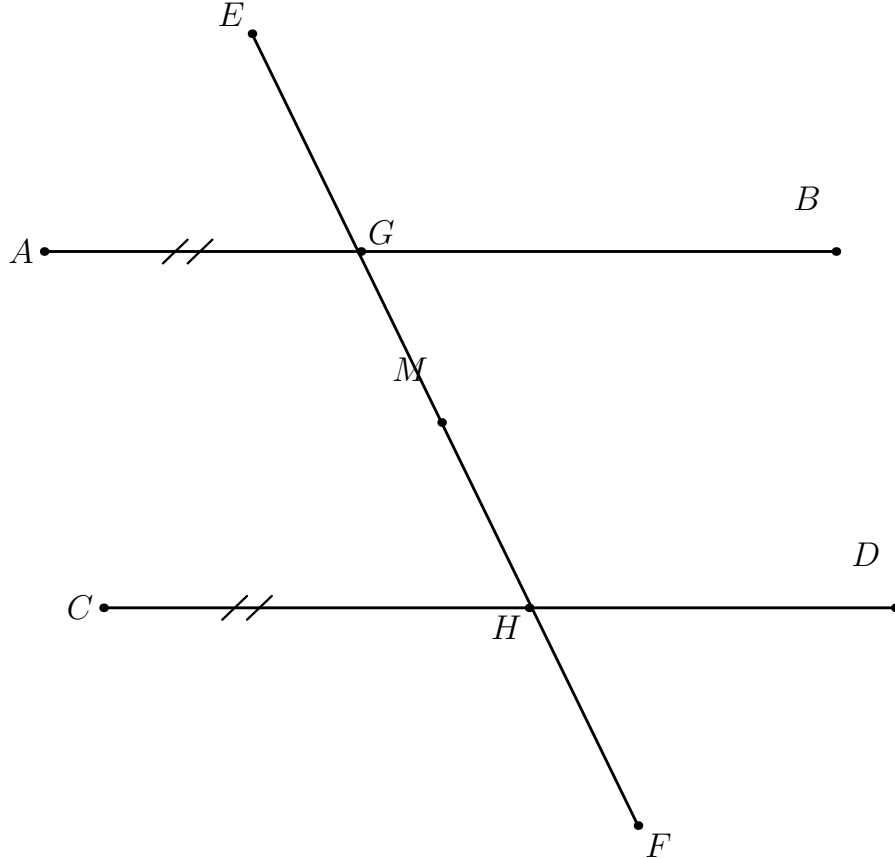


Figure 2: The formal transversal state used by `euclid_proposition_29_transversal`. The hidden witness M satisfies $G - M - H$. The selected points A and D are required to lie on opposite sides of the transversal. The point M provides the two same-ray normalizations used to pass from the line-level alternate-angle theorem to the endpoint formulation of I.29.

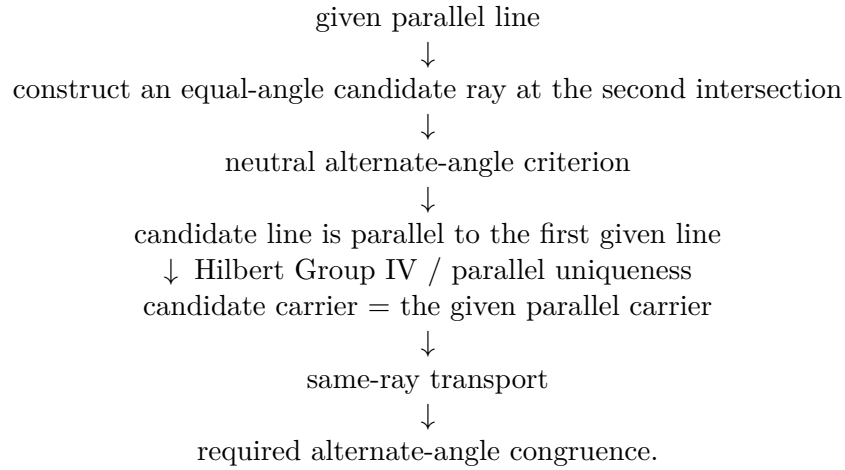
5 Where Group IV Enters

The Euclidean step is inside the reusable theorem

```
hilbert_alternate_angles_of_parallel_oppositeSide_lines
```

not after that theorem has returned.

Its internal structure is:



This existence–uniqueness decomposition is important. Neutral geometry can construct a line through the second intersection with the desired angle and can prove it parallel by the I.27 direction. The Euclidean axiom is needed to identify that constructed parallel with the already given parallel line.

The lower-level theorem constructs a candidate endpoint on the required ray. The final proof establishes that this endpoint and the already supplied endpoint determine the same ray. It does not need to prove that the two points are distinct. Therefore no separate diagram is drawn with two visibly distinct candidate endpoints: such a picture would assert more geometry than the theorem provides.

6 The Two Neutral Corollary Branches

6.1 Corresponding angles

The theorem

`euclid_proposition_29_corresponding`

first calls `euclid_proposition_29_transversal`. It then uses the order data $A - G - B$ and $E - G - H$ to obtain the vertical-angle congruence

$$\angle EGB \cong \angle AGH.$$

Transitivity with the alternate-angle result gives

$$\angle EGB \cong \angle GHD.$$

No new point, line, intersection, or side relation is introduced here.

6.2 Same-side interior angles

The theorem

`euclid_proposition_29_same_side`

again reuses the alternate-angle theorem. The order relation $A-G-B$ provides the supplementary ray to GB , and the result is packaged in the synthetic predicate

`HilbertAnglesEqualTwoRightAnglesWithSupplement`

rather than in a numerical angle-sum language.

Finally,

`euclid_proposition_29_all`

packages the alternate, corresponding, and same-side conclusions. It adds no new geometry.

7 Formal Dependency Graph

At the level of the current production file the dependency structure is

$$\begin{array}{c} \text{euclid_proposition_29} \\ \downarrow \\ \text{equal_angles_from_parallel} \\ \downarrow \\ \text{Hilbert Euclidean alternate-angle theorem} \end{array}$$

and, for the explicit transversal API,

$$\begin{array}{c} AB \parallel CD \\ \downarrow \\ GA \parallel HD \\ \downarrow \\ G - M - H \\ \downarrow \\ \text{hilbert_alternate_angles_of_parallel_oppositeSide_lines} \\ \downarrow \\ \angle MGA \cong \angle MHD \\ \downarrow \text{ same-ray normalization} \\ \angle AGH \cong \angle GHD \end{array}$$

followed by two neutral branches:

$$\begin{array}{ccc} & \angle AGH \cong \angle GHD & \\ \swarrow & & \searrow \\ \text{vertical angles} & & \text{supplementary configuration} \\ \downarrow & & \downarrow \\ \angle EGB \cong \angle GHD & & \text{same-side interior angles equal } 2R \text{ synthetically.} \end{array}$$

There is no geometric case split in `Proposition29.lean`.

8 Diagrammatic Audit

The preserved Book1R chapter used three figures. All three showed the same basic geometry. The first recorded the transversal configuration, while the second and third only changed which angle arcs were highlighted.

Under the present audit rule, those are not three geometric states. The final documentation therefore uses only two figures:

1. the classical transversal configuration, shared by all three of Euclid’s conclusions;
2. the formal transversal-normalization state with the hidden witness M and the explicit opposite-side orientation required by the Lean API.

The old separate alternate-angle and corresponding-angle pictures are not retained as separate floats because no new point, construction, intersection, or order/side configuration appears between them. They record a change of logical interpretation, not a change of geometry.

The deeper candidate-ray witness used inside the reusable Hilbert theorem is also not drawn separately. The proof ultimately establishes only that its ray agrees with the given target ray; drawing two distinct candidate points would add an unproved inequality.

9 Logical Position of Proposition I.29

The block I.27–I.29 has a clean logical boundary:

- I.27 : equal alternate angles $\longrightarrow$ parallel lines,
- I.28 : other angle conditions $\longrightarrow$ parallel lines,
- I.29 : parallel lines $\longrightarrow$ angle relations.

The first two directions are neutral. The third requires

[HilbertEuclideanPlane Geo]

and hence the project-level form of Hilbert’s Group IV.

This explains why I.29 becomes a central transport theorem for the next part of Book I: it feeds I.30, supplies the uniqueness side of the discussion around I.31, is used in I.32, and provides the parallel-to-angle interface used in later parallelogram propositions.

10 Conclusion

Proposition I.29 is the first genuinely Euclidean parallel-to-angle theorem in the Book I reconstruction. Its formal core is the alternate-angle implication. The current reusable Hilbert proof exposes the exact role of Group IV: neutral geometry constructs a candidate parallel direction, while parallel uniqueness identifies that direction with the line already present in the hypotheses. The explicit Book I theorem then uses the hidden point M only to normalize the line-level result along the transversal.

The corresponding-angle and same-side conclusions introduce no additional geometry. For that reason the audited chapter needs two figures rather than three: one for Euclid's common transversal configuration and one for the formal hidden geometry actually present in the Lean proof.